Darius Dayu Giannoli

darius.giannoli@epfl.ch | +33 6 72 16 26 63 | linkedin.com/in/darius-giannoli | github.com/DariusGiannoli

Research interests: Wearable Robotics, Vision-Based Control, Human-Robot Interaction, Biomedical Robotics, Haptics.

Education

Swiss Federal Institute of Technology in Lausanne (EPFL) – BSc, Computer Science Expected Graduation : 2027

- Relevant Coursework: Object Oriented Programming, Numerical Methods, Machine Learning, Electronics

Brussels French School Jean-Monnet, French Science Baccalaureate, European Section Graduation Date : 2023

- Graduated with highest honors with jury recognition (top 1% nationwide)

Research Experience

Humanoid Robotics Lab (prof Xie Ming), NTU - Global COnnect Fellowship Feb 2026 - March 2026

Laboratory of Intelligent Systems (Prof. Floreano), EPFL - Undergraduate Researcher Jun 2025 - Present
Full-time Summer Intern (Jun – Sep 2025) → Extended to semester research (Sep 2025 – Present)

- Researching on haptic feedback for drone-swarm embodiment & maximizing multi-modal feedback
- Developing torso-and-arms wearable device combining vibration and compression, with real-time control
- Conducting behavioral studies to maximize parallel information throughput leveraging multi-channel encoding
- Working on real-time priority based orchestrating algorithms for simultaneous pattern representations

Experience

ML, Haptics & Software Engineer, *EPFL N-pulse make-project* Sept 2024 – Present

- Leading development of force, texture, and proprioception feedback for haptic gloves and prosthesis
- Leading development of vision and audio based humanoid robotics/prosthesis complete grasping pipeline
- Developed a pipeline for prosthesis socket adaptation with automated electrode positioning (Scikit-learn, Open3d)

Telecommunications Engineer, *EPFL Spacecraft Team make-project* Sept 2024 – June 2025

- Worked on HMAC algorithm for telemetry telecommunications
- Worked on the Ultra High Frequency (UHF) antenna and transceiver and PC104 for communication tests

Data Intern, *LC3 Project, EPFL Laboratory of Construction Materials (LCM)* July - August 2024

- Lead the development of a database & sample tracking solution while being in charge of another intern
- Streamlined data management and accessibility & processed large datasets for accuracy and consistency

Leadership & Honors

Laidlaw Leadership and Research Scholar, Laidlaw Foundation 2025 – Present

- Highly selective scholarship (25 students) combining funded research internship with leadership development

Team Leader, Software & Haptics, EPFL Npulse Make Association Sept 2025 - Present

- Managing 12 MSc students; in charge roadmap/OKRs, timeline, weekly meetings & tasks integration
- Supervising development of Vision/EMG/EEG signal merging, firmware, Simulation, Mobile App, Scanning projects

Rugby Captain & Player, Boitsfort Rugby Club – Belgium 2011 – 2022

- Competitive player & Captain (2018-2022): 4x Belgian national champion, represented Belgium internationally

Skills & Projects

ProstGrasp — Vision & Audio based grapsing with Haptic feedback pipeline for prosthesis (EPFL Npulse)

- Audio & Vision based grasping Pipeline using Point Cloud, Computer Vision, Imitation & Reinforcement learning
- Simple prosthesis based on HuggingFace Lerobot, controlled by Vision Pipeline & integrated with Haptic Feedback

VibraForge — Haptic Pattern Designer (EPFL LIS- Utoronto)

- GUI (PyQT6) toolkit for designing/AB-testing patterns (incl. real time phantom routing); allowing real-time scheduling
- Serial/BLE/ESP-NOW communication protocol with optimized packet creation & hardware processing

Programming Languages: Java, Python, C/C++, JavaScript, HTML/CSS

Technologies: Github, Pytorch, Numpy, Flask, Open3D, Pandas, MPLAB X IDE, Arduino IDE
Languages: French (Native), English (C1/C2), Spanish(B2)